EngSci 721

Inverse Problems and Learning From Data

Oliver Maclaren (oliver.maclaren@auckland.ac.nz)

Module overview

Inverse Problems and Learning From Data (*Oliver Maclaren*)

[~14 lectures/3 tutorials]

1. *Basic concepts* [5 lectures + 1 Tutorial]

Forward vs inverse problems. Well-posed vs ill-posed problems. Algebra and calculus of inverse problems (left and right inverses, generalised and pseudo inverses, resolution operators, matrix calculus). Representing higher dimensional problems (image data etc).

2. *Instability and regularisation* [6 lectures + 1 Tutorial]

Instability and related issues for generalised inverses. Introduction to regularisation and trade-offs. Tikhonov regularisation. Higher-order Tikhonov regularisation. Sparsity and regularisation using different norms. Truncated singular value decomposition. Iterative regularisation, including stochastic/mini-batch gradient descent.

3. *Further topics* [3 lectures + 1 Tutorial]

Regularisation parameter choice, including statistical and machine learning views of regularisation. Confidence sets for linear and nonlinear models. Physics-informed machine learning and neural networks.

Lecture 1: Overview

Topics:

- Forward vs inverse problems and learning from data
- Examples
- Well-posed vs ill-posed problems
- Illustrations of ill-posedness

Housekeeping

Assessment

- Two assignments — 10% each
  A1: Mon 14 Sep, 11:59 pm; A2: Fri 23 Oct, 11:59 pm
- Two in-class tests — 15% each
  Test 1: Fri 21 Aug; Test 2: Fri 16 Oct
- Final exam — 50%

Weekly classes

Day	Time	Room
Wed	14:00–15:00	OGGB 055
Thu	15:00–16:00	OGGB 057
Fri	15:00–16:00	Arch. West 301

Same timetable throughout the semester. Oliver: Weeks 1–6; Ru: Weeks 7–12. Full schedule and updates are on Canvas.

EngSci 721 : Inverse Problems & 'Learning from data'

Lecture 1: Overview.

- Inverse problems, UQ, data etc
- Forward vs inverse problems
- Examples of inverse problems
- Well-posed & ill-posed problems
- Illustrations of ill-posedness

Prelude: Models + Data.

Inverse Problems, UQ, Data-centric engineering, scientific machine learning...

The imperative of physics-based modeling and inverse theory in computational science

To best learn from data about large-scale complex systems, physics-based models representing the laws of nature must be integrated into the learning process. Inverse theory provides a crucial perspective for addressing the challenges of ill-posedness, uncertainty, nonlinearity and under-sampling.

Karen E. Willcox, Omar Ghattas and Patrick Heimbach

The notions of 'artificial intelligence (AI) for science' and 'scientific machine learning' (SciML) are gaining widespread attention in the scientific community. These initiatives target development and adoption of AI approaches in scientific and engineering fields with the goal of accelerating research and development breakthroughs in energy, basic science, engineering, medicine and national security. For the past six decades, these fields have been advanced through the synergistic and principled use of theory, experiments and physics-based simulations. Our increased ability to sense and acquire data is clearly a game-changer in these endeavors. Yet, in our excitement to define a new generation of data-centric approaches, we must be careful not to chart our course based entirely on the successes of data science and machine learning in the vastly different domains of social media, online entertainment, online retail, image recognition, machine translation and natural language processing — domains for which data are plentiful and physics-based models do not exist. In contrast, many of today's scientific grand challenges suffer from the lack of adequate sampling of the processes underlying the complex, large-scale systems. Yet, for many of these systems, a great deal is known regarding the underlying physical principles or governing equations; we must continue to appeal to computational science to unleash this information. As Covsney et al. argue elegantly[1], big data need big theory — and big physics-based simulation models — too.

The unreasonable effectiveness of physics-based models

But what are physics-based models and why are they indispensable? A physics-based model is a representation of the governing laws of nature that innately embeds the concepts of time, space, causality and generalizability. These laws of nature define how physical, chemical, biological and geological processes evolve. Physics-based models typically encode knowledge in the form of conservation and constitutive laws, often based on decades if not centuries of theoretical development and experimental validation. These laws often manifest as systems of differential equations that are solved numerically with high-performance computing (HPC).

In his famous 1960 article, Eugene Wigner wrote about "The unreasonable effectiveness of mathematics in the natural sciences"[2], pointing to "the 'laws of nature' being of almost fantastic accuracy but of strictly limited scope." As Wigner discusses, physics-based modeling is powerful and effective because it gives us a predictive window into the future based on understanding. It achieves this because any particular model limits its scope to a particular class of physical systems or processes, building a universal representation within that class. Armed with that universal representation, physics-based modeling is a way to simulate 'what if' scenarios and to issue predictions that have explanatory power or projections with quantified uncertainties that go beyond the current state and available data. For example, in our modern world, physics-based models are used to issue predictions about the future evolution of a cancer patient's tumor, or about the loads that a yet to be built aircraft may find itself experiencing under different operating conditions. They enable predictions of weather over the next five to ten days, or scenario-based projections about the future state of the Earth's climate in the decades to come.

The role of inverse theory in learning from data

As attention turns from simulation to learning from data (that is, from the forward problem to the inverse problem), we must bring these learned lessons — the big theory and the big physics-based simulation models — with us. Without physical constraints, purely data-driven approaches are unlikely to be predictive, no matter how expressive the underlying representation. Even when physical models are not well-established (such as for many biological processes, in constitutive laws for complex materials, or in subgrid scale models for unresolved physics), we know that certain universal properties and relationships must hold, such as conservation properties, material frame indifference, objectivity, symmetries, or other invariants. The learning-from-data problem is fundamentally an inverse problem that merges the partial knowledge reservoir of data with that of physics-based models in a systematic and rigorous way, and in a way that exploits the complementary and mutually reinforcing aspects of both data and models.

Data and models invariably come with uncertainties. Data are often noisy, sparsely and heterogeneously sampled, and representative of disparate observables. Experiments and data gathering are costly, time-consuming, and sometimes dangerous or impossible. Often data are hardest to acquire and are thus sparsest in the most decision-critical regions (for example, failure, instability, extreme environments). Even if it is possible to generate more data (for example, via simulation) a fundamental challenge remains: due to information loss in the forward problem and resulting ill-posedness of the inverse problem, data often contain only low-dimensional information about the physics, even when the data are large-scale[3].

In turn, physics-based models are typically characterized by uncertain parameters, which may include initial and boundary conditions, sources, material properties, geometry and model structure, all of which can be heterogeneous in space or time. In this setting, rather than ignore known physics, we must employ them to define the maps from parameters to observables, and invert them to project

NATURE COMPUTATIONAL SCIENCE | VOL 1 | MARCH 2021 | 166–168 | www.nature.com/natcomputsci

2021

Prelude : Just use 'AI' ?

Physics-informed machine learning

George Em Karniadakis[1,2], Ioannis G. Kevrekidis[3,4], Lu Lu[5], Paris Perdikaris[6], Sifan Wang[7] and Liu Yang[1]

Abstract | Despite great progress in simulating multiphysics problems using the numerical discretization of partial differential equations (PDEs), one still cannot seamlessly incorporate noisy data into existing algorithms, mesh generation remains complex, and high-dimensional problems governed by parameterized PDEs cannot be tackled. Moreover, solving inverse problems with hidden physics is often prohibitively expensive and requires different formulations and elaborate computer codes. Machine learning has emerged as a promising alternative, but training deep neural networks requires big data, not always available for scientific problems. Instead, such networks can be trained from additional information obtained by enforcing the physical laws (for example, at random points in the continuous space-time domain). Such physics-informed learning integrates (noisy) data and mathematical models, and implements them through neural networks or other kernel-based regression networks. Moreover, it may be possible to design specialized network architectures that automatically satisfy some of the physical invariants for better accuracy, faster training and improved generalization. Here, we review some of the prevailing trends in embedding physics into machine learning, present some of the current capabilities and limitations and discuss diverse applications of physics-informed learning both for forward and inverse problems, including discovering hidden physics and tackling high-dimensional problems.

Simulation models + data

→ physics + ML?

Analysis https://doi.org/10.1038/s42256-024-00897-5

Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations

Received: 23 August 2023
Accepted: 13 August 2024
Published online: 25 September 2024
Check for updates

Nick McGreivy & Ammar Hakim

One of the most promising applications of machine learning in computational physics is to accelerate the solution of partial differential equations (PDEs). The key objective of machine-learning-based PDE solvers is to output a sufficiently accurate solution faster than standard numerical methods, which are used as a baseline comparison. We first perform a systematic review of the ML-for-PDE-solving literature. Out of all of the articles that report using ML to solve a fluid-related PDE and claim to outperform a standard numerical method, we determine that 79% (60/76) make a comparison with a weak baseline. Second, we find evidence that reporting biases are widespread, especially outcome reporting and publication biases. We conclude that ML-for-PDE-solving research is overoptimistic: weak baselines lead to overly positive results, while reporting biases lead to under-reporting of negative results. To a large extent, these issues seem to be caused by factors similar to those of past reproducibility crises: researcher degrees of freedom and a bias towards positive results. We call for bottom-up cultural changes to minimize biased reporting as well as top-down structural reforms to reduce perverse incentives for doing so.

Weak baselines, over hyped results?

What Has a Foundation Model Found? Inductive Bias Reveals World Models

Keyon Vafa [1] Peter G. Chang [2] Ashesh Rambachan [2] Sendhil Mullainathan [2]

Abstract

Foundation models are premised on the idea that sequence prediction can uncover deeper domain understanding, much like how Kepler's predictions of planetary motion later led to the discovery of Newtonian mechanics. However, evaluating whether these models truly capture deeper structure remains a challenge. We develop a technique for evaluating foundation models that examines how they adapt to synthetic datasets generated from some postulated world model. Our technique measures whether the foundation model's inductive bias aligns with the world model, and so we refer to it as an *inductive bias probe*. Across multiple domains, we find that foundation models can excel at their training tasks yet fail to develop inductive biases towards the underlying world model when adapted to new tasks. We particularly find that foundation models trained on orbital trajectories consistently fail to apply Newtonian mechanics when adapted to new physics tasks. Further analysis reveals that these models behave as if they develop task-specific heuristics that fail to generalize.

This path — from predicting sequences to understanding the deeper mechanisms that underlie them — is not unique to physics. In biology, animal breeders noticed patterns in the traits of offspring long before their predictive insight inspired Mendel to develop a theory of genetics.

How would we know if foundation models have also made the leap from making accurate predictions to developing reliable world models? This paper develops a framework for answering this question. Specifically, we create a procedure that, when given a foundation model and world model, tests whether the foundation model has learned that world model. We call this technique an *inductive bias probe*, and it is built on a simple insight: the implicit world model of a foundation model is revealed by how it extrapolates from a small amount of information. This is inspired by how scientists use world models — to make inferences from small amounts of data. Similarly, the inductive bias of a foundation model reveals its world model.

We first demonstrate this procedure using an example from physics. Specifically, we aim to replicate Kepler's and Newton's experiments, albeit replacing the physicist with a foundation model of orbital mechanics. Much like Kepler, the model is able to predict orbital trajectories, even for solar systems it has not seen.

prediction on specific task ≠ general understanding (AI vs Newton)

Statistics? (Frequentist? Bayesian? ...)

Inverse problems as statistics

Steven N Evans[1] and Philip B Stark[2]

[1] Department of Statistics and Department of Mathematics, University of California, Berkeley, CA 94720-3860, USA
[2] Department of Statistics, Space Sciences Laboratory, and Center for Theoretical Astrophysics, University of California, Berkeley, CA 94720-3860, USA

E-mail: evans@stat.berkeley.edu and stark@stat.berkeley.edu

Received 27 September 2001, in final form 26 February 2002
Published 20 May 2002
Online at stacks.iop.org/IP/18/R55

Abstract

What mathematicians, scientists, engineers and statisticians mean by 'inverse problem' differs. For a statistician, an inverse problem is an inference or estimation problem. The data are finite in number and contain errors, as they do in classical estimation or inference problems, and the unknown typically is infinite dimensional, as it is in nonparametric regression. The additional complication in an inverse problem is that the data are only indirectly related to the unknown. Canonical abstract formulations of statistical estimation problems subsume this complication by allowing probability distributions to be indexed in more-or-less arbitrary ways by parameters, which can be infinite dimensional. Standard statistical concepts, questions and considerations such as bias, variance, mean-squared error, identifiability, consistency, efficiency and various forms of optimality apply to inverse problems. This paper discusses inverse problems as statistical estimation and inference problems, and points to the literature for a variety of techniques and results. It shows how statistical measures of performance apply to techniques used in practical inverse problems, such as regularization, maximum penalized likelihood, Bayes estimation and the Backus–Gilbert method. The paper generalizes results of Backus and Gilbert characterizing parameters in inverse problems that can be estimated with finite bias. It also establishes general conditions under which parameters in inverse problems can be estimated consistently.

Motivation / Working Principle:

→ The mathematics underlying 'classical' inverse problems sheds light on statistics, machine learning etc as well as vice-versa

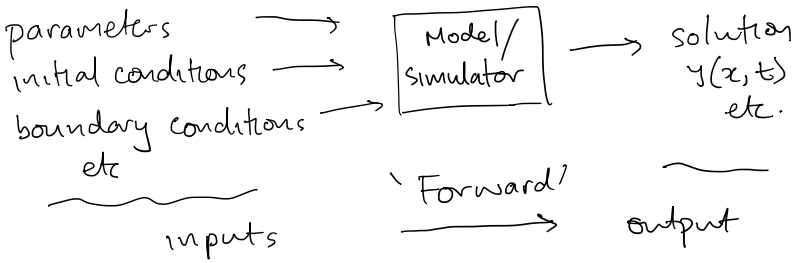
} we'll look at both!

statistics : mappings from data to parameter
ML : mappings from data to data
} eg

inverse problems : mappings between general spaces

Forward vs Inverse Problems

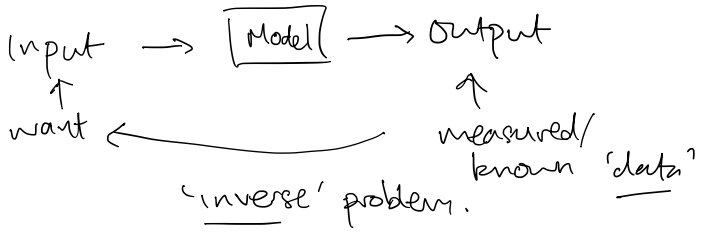
Typical modelling/simulation setup:



& Typical 'forward' models
- ODE
- PDE
- ABM (Agent-based model
- etc
- (Many things!)

Problem: usually measure (noisy) outputs (data!) & want inputs

ie



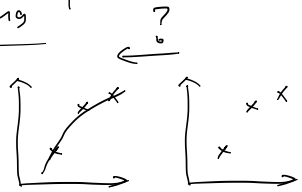
Examples

Many important problems can be considered 'inverse' problems

Obvious example:

parameter estimation, fitting, parameterisation, learning

eg 'fit' an ODE, PDE, polynomial, neural net etc to given data.



But the general concept is very broad & essentially about (math) 'inverting' given functions, or (physics) determining 'causes' from 'effects'

eg $y = f(x)$, say $y = x^2$

$x \longmapsto x^2$

observe $y = 4$

what was x?

Note: no unique sol^n!

↳ solution set: $\{2, -2\}$

(note: $f^{-1}\{4\}$, not $f^{-1}(4)$!)

Examples (see readings for details)

Some common 'physical' examples include

- Tomography — CT (computerised), EIT (electrical impedance) } etc
  - 'imaging by sections'
    - → given slices, reconstruct object
- Deconvolution (deblurring)
  - given a blurred photograph, reconstruct unblurred version
- Geophysics — determine the origin of an earthquake, given seismographs measuring arrival times
- Geothermal engineering — determine the density or permeability etc of underground rocks given surface or well measurements of temperature

Examples (see Aster et al L1 reading for details)

More physical/mathematical examples:

- Determine the origin of a groundwater contaminant after it has been transported (by advection/diffusion) downstream to measurement locations
- Given a function, determine its derivative!
  - HUH? See later →
- Given crime scene clues, determine the perpetrator....

→ many more!

What's the difference between 'forward' & 'inverse' though?

Keller (1976): two problems are inverse to each other if the formulation of one is naturally 'opposite' to the other

→ eg differentiation & integration
→ intuitive rather than fully-formal

However, often one problem is more naturally considered the direct or forward problem & the other more naturally the inverse problem

---

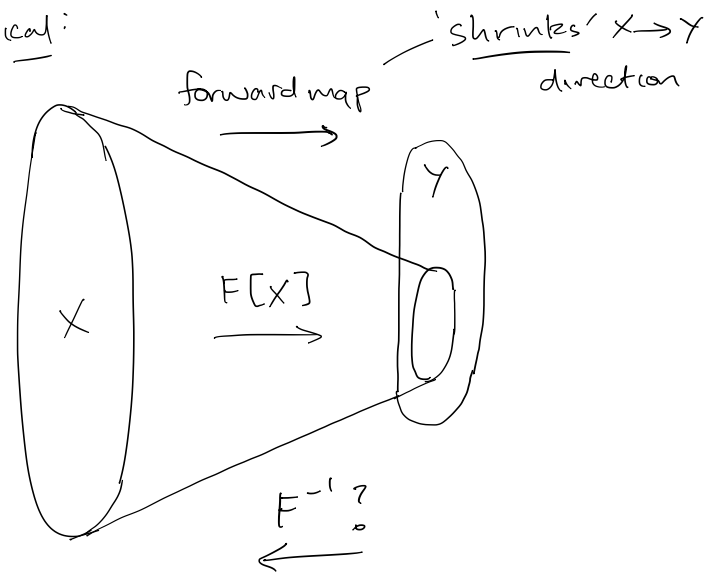
Example pairs:

Forward	Inverse
multiplication to find product	division to find factors
given present/past predict future	given present/future predict past
integration	differentiation
↓	↓
∘ 'easier'	∘ 'harder'
∘ well-defined	∘ can be ill-defined eg $\frac{1}{0}$
∘ tend to 'smooth', 'combine' or 'reduce' inputs to simpler output	∘ often require recovering 'lost' info or 'uncombining' to get inputs from combined output
∘ respect causality/ flow of time	∘ amplify noise
∘ stable	∘ acausal/backwards in time?
	∘ unstable?

features

Mathematical Picture

Typical:

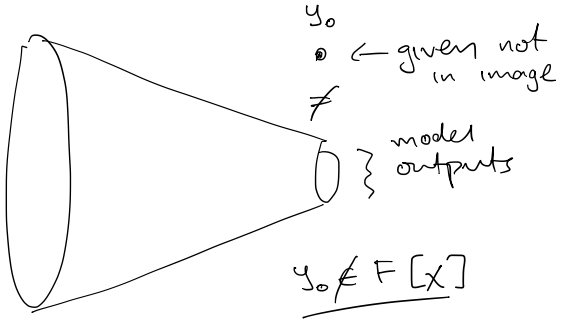


Think:

size {inputs/domain space} > size {outputs/range/image space}

---

also:



$y_0 \notin F[X]$

Well-posed vs ill-posed Problems

Hadamard (1902) call a problem well-posed if a solution:

- Exists (possible to satisfy)
- Is unique
- Is stable wrt 'small' changes in the givens (continuity)

Problems that are not well-posed are said to be

ill-posed

Key: forward problems → well-posed
inverse problems → ill-posed
(generally speaking)

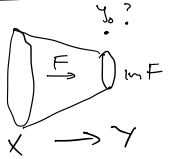
Well-posed vs ill-posed Problems

Math: given model $F: X \to Y$ & data $y_0 \in Y$

we are essentially looking for the inverse F^{-1}: when does it exist? when unique? when continuous/stable

---

1. **Existence**: $y_0 \in \text{im}F$ (F is onto)



$\text{im}F$ = 'image of F'
$= F[X]$
$= \{F(x) \mid x \in X\}$

note: image of a set is a set.

---

2. **Uniqueness**: (F is 1–1)

$F(x_1) = F(x_2) \Rightarrow x_1 = x_2$

ie

$x_1 \neq x_2 \Rightarrow F(x_1) \neq F(x_2)$



NO! Want: distinct → distinct.

---

3. **Stability**: F^{-1} (which exists if 1 & 2 sat.) is (sufficiently) continuous:

if $y_\delta \approx y$ then $F^{-1}(y_\delta) \approx F^{-1}(y) = x$

y_δ: small data perturb; $F^{-1}(y_\delta)$: small soln perturb.

Stability/continuity can be more formally defined in various ways.

- **Sequential continuity of F^{-1}**: for any sequence $y_n \to y$ & $x = F^{-1}y$ we have $F^{-1}(y_n) \to F^{-1}(y) = x$

(F^{-1} (converging sequence of 'input' data) = converging sequence of 'output' soln's)

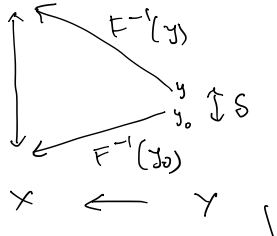
- **ϵ-δ defn's**

$(\forall \epsilon)(\exists \delta(\epsilon))\left[\|y - y_0\| < \delta \Rightarrow \|F^{-1}(y) - F^{-1}(y_0)\| < \epsilon\right]$

[Given a tolerance ϵ, find a δ for that tolerance, s.t. if 'inputs' of F^{-1} differ by $< \delta$ then 'outputs' of F^{-1} differ less than ϵ.]

practical/sufficient: can't take δ lower than some 'resolution'

control via δ →

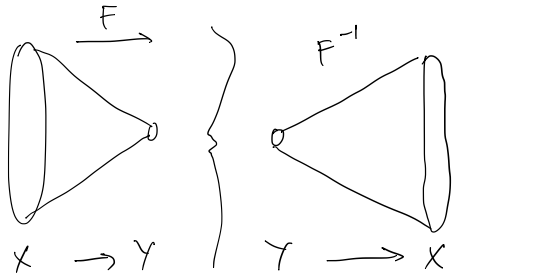


Well-posed vs ill-posed Problems

Intuition:

- Forward 'shrinks' (eg 'compact operator')
- Inverse needs to 'expand'

→ risks instability



small change → large change in inverse.

Key Message:

Even when inverse exists in principle it may be unstable

(might be 1-1 but 'close to' non 1-1)

Illustrations: instability even when inverse exists

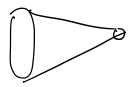
Integration: 'sums up' → reduces

eg $1+2+3 = 6$

$3+3 = 6$

$2+4 = 6$

&



differentiation: 'breaks down' → expands?

↳ Numerical Example.

- Given a signal 'vector' $x \in \mathbb{R}^n$,
- We can sum the first k elements by taking dot product with

$w_k = (1\ 1\ 1\ \cdots\ 1\ 0\ 0\ \cdots\ 0)^T = (1_k^T\ 0\ 0\ \cdots\ 0)^T$ (k ones)

$1_k^T = (1, 1, \cdots, 1)$ (k ones)

(Note we will often think of 'samples'/'signals' as finite dimensional vectors $(x_1, x_2, \cdots)$ approximating infinite-dimensional random vars $X(\Omega)$, $\Omega \in \mathcal{\Omega}$, & operations like summation etc as finite versions of integration).

Illustrations

Note: (assume all vectors are column by default)

Eg $x = (1 \;\; 2 \;\; 3 \;\; 0 \;\; 0)^T$

$w_1 = (1 \;\; 0 \;\; 0 \;\; 0 \;\; 0)^T$

$w_2 = (1 \;\; 1 \;\; 0 \;\; 0 \;\; 0)^T$

$\vdots$

$\text{cumsum}(k, x) = w_k^T x$

eg $cs(2, x) = (1 \;\; 1 \;\; 0 \;\; 0 \;\; 0) \begin{pmatrix} 1 \\ 2 \\ 3 \\ 0 \\ 0 \end{pmatrix}$

$= 1 + 2 = 3$

Return vector of all cumulative sums

$$A = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ \vdots & & & & \\ 1 & 1 & \cdots & & 1 \end{pmatrix} \leftarrow \text{forward map } X \to Y$$

$y = Ax$

Illustrations

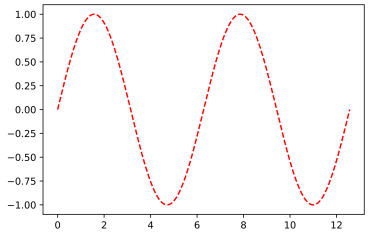
(import numpy as np)

Python (see Canvas)

```
t = np.linspace(0,4*np.pi,1000)
x = np.sin(t)
plt.plot(t,x,'r--')
plt.show()
```

→

Input signal. $x(t)$

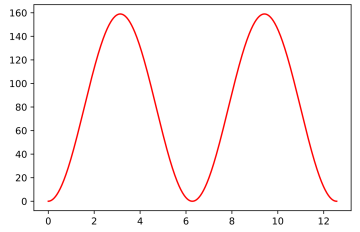


numpy's 'cumsum'

```
#built-in 'integration'
plt.plot(t,np.cumsum(x),'r')
plt.show()
```

→

'Integral' $y = \int_0^t x(\xi) d\xi$



Roll our own as matrix mapping

```
#forward mapping for integration (cumulative sum)
def create_fmap_int(d=1):
    A = np.zeros([d,d])
    for i in range(0,d):
        A[i,:i+1] = 1
    return A


#create forward mapping
A = create_fmap_int(len(x))
print(A)
```

→

```
[[ 1.  0.  0. ...,  0.  0.  0.]
 [ 1.  1.  0. ...,  0.  0.  0.]
 [ 1.  1.  1. ...,  0.  0.  0.]
 ...,
 [ 1.  1.  1. ...,  1.  0.  0.]
 [ 1.  1.  1. ...,  1.  1.  0.]
 [ 1.  1.  1. ...,  1.  1.  1.]]
```

---

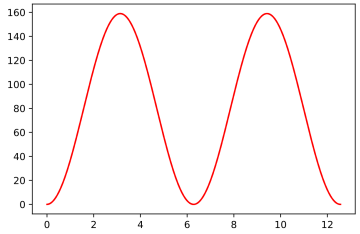
check A correct

```
#calculate output of forward mapping
y = np.dot(A,x)

#compare our forward mapping to built-in
plt.plot(t,np.cumsum(x),'r--')
plt.plot(t,y,'r')
plt.show()
```

→

indistinguishable output

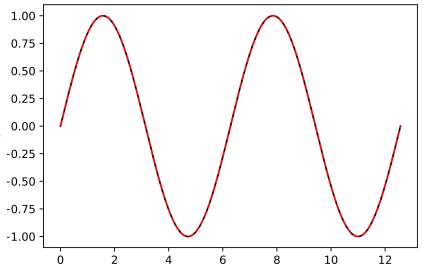


Invert using A^{-1} & compare to original output

```
#invert noise-free case
plt.plot(t,np.dot(np.linalg.inv(A),y),'k')
plt.plot(t,x,'r--')
plt.show()
```

→

indistinguishable input



Note: A^{-1} exists (1-1 & onto)

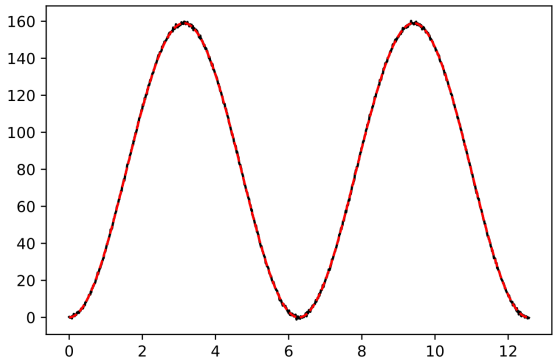
→ A is square
& all rows/cols are linearly independent

stability?

Add small amount of noise to output

ie $y_{observed} = Ax + \epsilon = y + \epsilon$

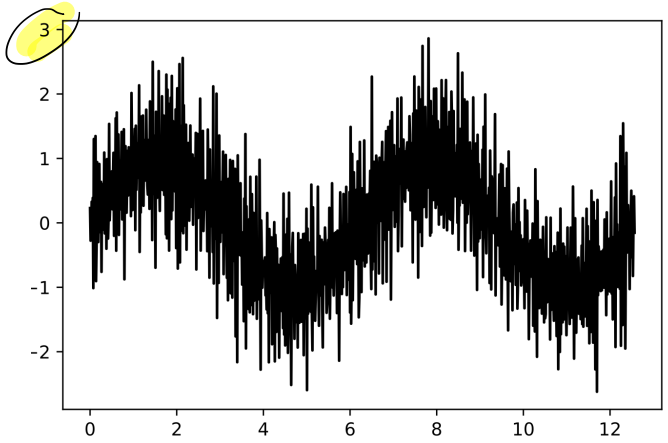
```
#add almost undetectable noise
y_noisy = y+np.random.normal(0,0.5,size=len(y))
plt.plot(t,y_noisy,'k')
plt.plot(t,y,'r--')
plt.show()
```



Question: is A^{-1} still 'good'?

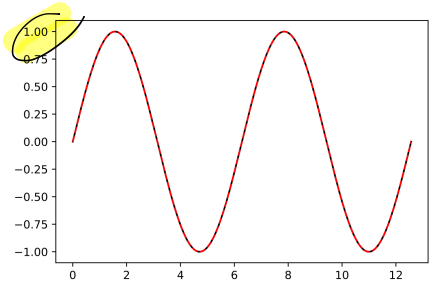
Instability of differentiation

```
#invert noisy
plt.plot(t,x,'r--')
plt.plot(t,np.dot(np.linalg.inv(A),y_noisy),'k')
plt.show()
```



Very 'irregular'! Large deviations

(Recall noise-free:



)

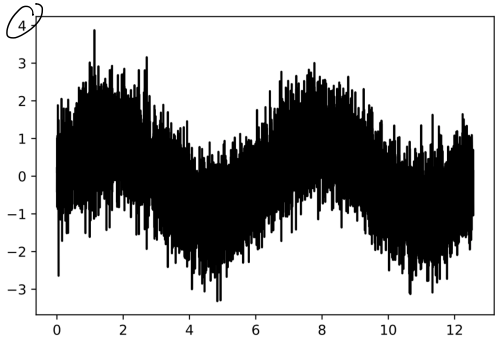
Can we just make grid finer?

→ Makes worse!

Finer grid: x 10 finer } like finer discretis.

```
t = np.linspace(0,4*np.pi,10000)
x = np.sin(t)
plt.plot(t,x,'r--')
plt.show()
```

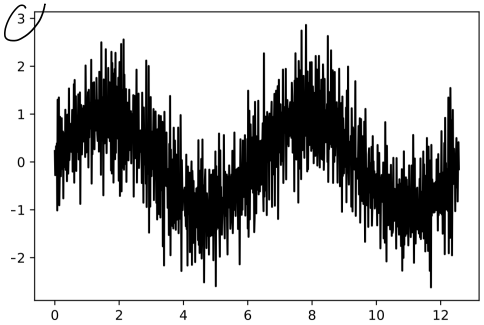
Same process gives:



noise ↑

even more!

compare:



Instability

- Integration 'smooths'
- Differentiation 'coarsens'

Mathematically, differentiation is an unbounded (discontinuous) operator [see functional analysis]

Numerically, differentiation is

ill-conditioned

↳ not quite unbounded since discrete / finite, but practically unbounded.

---

discretise ↓ continuous : ill-posed
discrete : ill-conditioned

---

see [Engel et al LI reading]

& →

Differentiation as discontinuous

Sequential continuity : f (seq.) continuous

means: $x_n \to x \Rightarrow f(x_n) \to f(x)$

Now consider sequence $y_n = \frac{1}{n}\sin(nx)$.

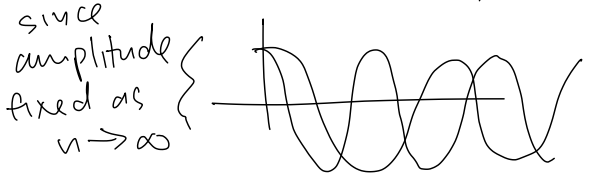
Now:

$n \to \infty$, $y_n \to 0$ (why ?)

And clearly $\frac{d}{dx}(0) = 0$

but $\frac{d}{dx}\left(\frac{1}{n}\sin(nx)\right) = \cos nx \not\to 0$

since amplitude fixed as $n \to \infty$



So not the case that for all y_n

$y_n \to y$ implies $F^{-1}(y_n) \to F^{-1}(y)$

(not 'A implies B' $\equiv$ A and not B)

$\Rightarrow F^{-1}$ is not (seq.) continuous.

Illustration : Local averaging (smoothing)

Given 'signal' or 'input' $x \in \mathbb{R}^n$

Local average over neighbourhood $w(i)$ of i is:

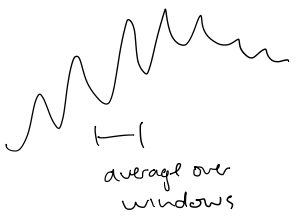
$$\frac{1}{|W|} \sum_{i \in W(i)} x(i)$$

$\cdots \quad x_{j-1} \; x_j \; x_{j+1} \; \cdots$

$\longleftrightarrow$

$w(j)$

Smoothing: moving average



average over windows



smoothed version.

Inverse Problem:

- given a smoothed (or 'blurred') image, determine original

→ eg photography

same idea as integration / differentiation

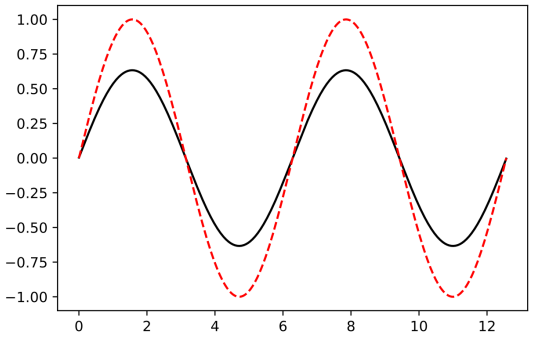
→ smoothing forward map

→ inverse amplifies any noise in output when trying to recover

$$y_0 = y + \epsilon = \underbrace{Ax}_{\text{smoothing}} + \underbrace{\epsilon}_{\text{noise}}$$

A^{-1} unstable!

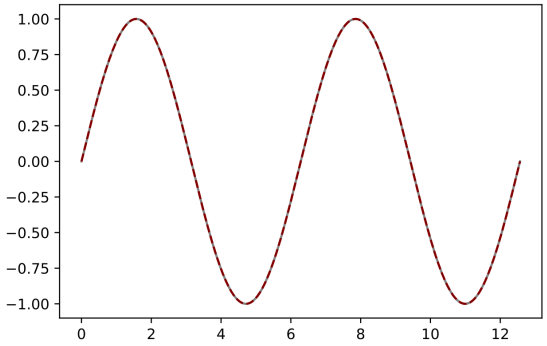
Smoothing map:



red: original
black: local av.

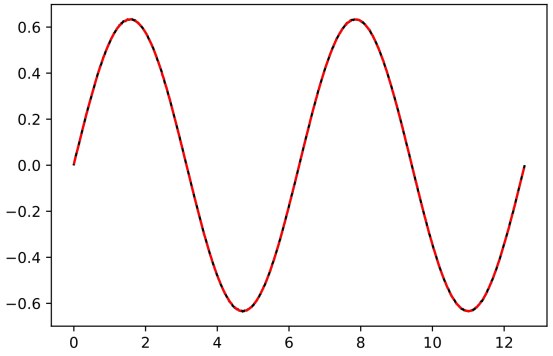
Can implement as linear forward map, ie matrix A (exercise?)

Invert noise free gives:



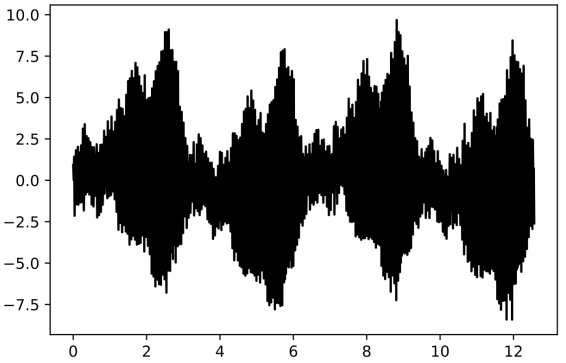
Instability of 'deblurring':

Add small noise:



practically indetectable!

Invert noisy using A^{-1}



Terrible!

Should we give up?

No! (Well...)

We can stabilise the inversion procedure

"Regularisation"

→ A key theme of the course!

We will first look at the existence & uniqueness of 'inverses', however, then at their stability

We will also develop matrix calculus along the way

↳ key to machine learning too!

Engl, Hanke, Neubauer et al 'Regularization of Inverse Problems'

1. Introduction: Examples of Inverse Problems

In this introductory section, we give, after a general discussion of the term "inverse problems", examples of various classes of inverse problems arising in various application fields. As we will see, *linear inverse problems* frequently lead to *integral equations of the first kind*, which is why such equations play an important role in the study of inverse problems, as we will see throughout a large part of this book. On the other hand, many basic *inverse problems* are inherently nonlinear even if the corresponding *direct problem* is linear. This fact is our motivation for devoting a significant part of this book to the mathematics of nonlinear inverse problems.

When using the term *inverse problem*, one immediately is tempted to ask "inverse to what?". Following J.B. Keller [150], one calls *two* problems *inverse to each other* if the formulation of one problem involves the other one. For mostly historic reasons, one might call one of these problems (usually the simpler one or the one which was studied earlier) the *direct problem*, the other one the *inverse problem*. However, if there is a *real-world* problem behind the mathematical problem studied, there is, in most cases, a quite natural distinction between the direct and the inverse problem. E.g., if one wants to predict the future behaviour of a physical system from knowledge of its present state and the physical laws (including concrete values of all relevant physical parameters), one will call this the *direct problem*. Possible *inverse problems* are the determination of the present state of the system from future observations (i.e., the calculation of the evolution of the system backwards in time) or the identification of physical parameters from observations of the evolution of the system (*parameter identification*).

There are, from the applications point of view, two different motivations for studying such inverse problems: first, one wants to *know* past states or parameters of a physical system. Second, one wants to find out how to influence a system via its present state or via parameters in order to *steer* it to a desired state in the future.

Thus, one might say the *inverse problems are concerned with determining causes for a desired or an observed effect.*

As we will see, such inverse problems most often do not fulfill Hadamard's postulates of *well-posedness* (see Chapter 2). They might not have a solution in the strict sense, solutions might not be unique and/or might not depend continuously on the data. Mathematical problems having these undesirable properties are called *ill-posed problems* (*improperly posed problems*) and pose (mostly because of the discontinuous dependence of solutions on the data) severe numerical difficulties. While the study of concrete inverse problems frequently involves the question how to enforce uniqueness by additional information or assumptions, not much can be said about this in a general context. The aspect of lack of stability and its restoration by appropriate methods (*regularization methods*), however, can be treated in sufficient generality. The theory of regularization methods is well-developed for linear inverse problems and at least emerging for nonlinear problems and forms the core of this

book.

There is a vast literature on inverse and ill-posed problems. In addition to the books already quoted in the Preface, we mention, as general references,

- the following monographs: [13, 32, 45, 95, 136, 225, 230, 282, 288],
- the following conference proceedings: [14, 36, 43, 72, 78],[81] (which emphasizes inverse problems arising in industry), [83, 138, 223, 241, 242, 265, 289],
- the journals *Inverse Problems* (Institute of Physics Publ.), *Inverse Problems in Engineering* (Gordon & Breach), and *Journal of Inverse and Ill-Posed Problems* (VSP).

Many more references will be given in the appropriate sections.

1.1. Differentiation as an Inverse Problem

Two mathematical problems inverse to each other are differentiation and integration. A-priori, it is not clear which of these problems should be the direct problem and which one the inverse problem. However, as we will now see, differentiation has (as opposed to integration) the properties of an ill-posed problem. Moreover, since, as we will see later, many inverse problems involve at some step differentiation of the data, differentiation might be viewed as the "inverse problem", although in most calculus courses, it is treated first.

Let $f \in C^1[0,1]$ be any function, $\delta \in (0,1)$, $n \in \mathbb{N}$ $(n \geq 2)$ be arbitrary, and define

$$f_n^\delta(x) := f(x) + \delta \sin \frac{nx}{\delta}, \quad x \in [0,1]. \tag{1.1}$$

Then

$$(f_n^\delta)'(x) = f'(x) + n \cos \frac{nx}{\delta}, \quad x \in [0,1]. \tag{1.2}$$

Now, in the uniform norm,

$$\|f - f_n^\delta\|_\infty = \delta,$$

but

$$\|f' - (f_n^\delta)'\|_\infty = n.$$

Hence, if we consider f and f_n^δ as the exact and perturbed data, respectively, then for an arbitrarily small data error δ, the error in the result, namely the derivative, can be arbitrarily large, namely n. Hence, the derivative does not depend continuously on the data with respect to the uniform norm. Of course, we could enforce continuous dependence by measuring the data error in the C^1-norm. However, this would be a sort of cheating, since then, we would call a data error small if the error in the function values and in the values of the derivative, which is exactly what we want to compute, would be small.

Engl, Hanke, Neubauer et al 'Regularization of Inverse Problems'

For later reference, note that f' solves the simple integral equation of the first kind

$$(Kx)(s) := \int_0^s x(t)\,dt = f(s) - f(0)\,, \tag{1.3}$$

which is solvable in $C[0,1]$ only if $f \in C^1[0,1]$. The corresponding direct problem would be to compute f from x, i.e., integration, which is a stable process on $C[0,1]$. Note that integration is a smoothing process, i.e., highly oscillatory errors in x (e.g., of the form $n\cos(nx/\delta)$ as they appeared in (1.2)) are damped out (to $\delta\sin(nx/\delta)$) and have a very small effect on the data for the inverse problem. This smoothing is responsible for the fact that errors of small amplitude, but high frequency, create large oscillations in the solution of the inverse problem. These considerations are not restricted to this concrete problem: whenever a direct problem has smoothing properties one has to expect the appearance of oscillations coming from small data perturbations (of high frequency) in the solution of the inverse problem. This effect is the more pronounced the stronger smoothing the direct problem is.

Why (or, under what circumstances) can we differentiate a function in spite of these problems? We have to be able to exclude the presence of data errors of arbitrarily high frequency, e.g., of the form as in (1.1); this can be done, e.g., if we know a bound for f''. In the example above, such a bound would give a bound for n in terms of δ, thus coupling the amplitude and the frequency of the possible data errors. A functional analytic argument is the following:

If we consider the operator K as defined in (1.3) on $C[0,1]$, then it is a continuous linear injective operator, whose inverse (defined on $C^1[0,1]$, considered as a subspace of $C[0,1]$) is unbounded. However, if we restrict K to the set $\{x \in C^1[0,1] \mid \|x\|_\infty + \|x'\|_\infty \le \gamma\}$, which is compact in $C[0,1]$ due to the Arzela-Ascoli Theorem, then the inverse of this restricted operator is continuous on its range, as the inverse of a continuous bijective (not necessarily linear) map defined on a compact set is again continuous. Thus, we can "restore stability" by assuming an a-priori bound for f' and f''.

The stability problems addressed must appear somehow when computing the derivative via difference quotients: let f be the function we want to differentiate, f^δ its noisy version with

$$\|f - f^\delta\|_\infty \le \delta\,.$$

We want to use the central difference quotient with step size h. If $f \in C^2[0,1]$, Taylor expansion yields

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + O(h)\,,$$

while for $f \in C^3[0,1]$,

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + O(h^2)\,.$$

Thus, the accuracy of the central difference quotient depends on the smoothness of

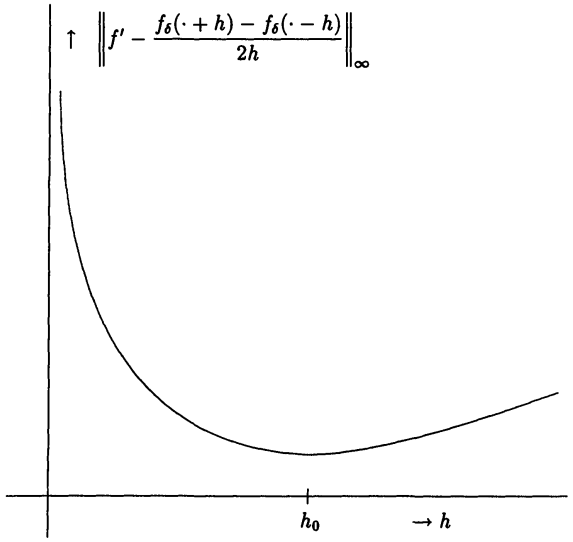


Figure 1.1: Total error depending on h

the exact data. Instead of f', we are actually computing

$$\frac{f^\delta(x+h) - f^\delta(x-h)}{2h} \sim \frac{f(x+h) - f(x-h)}{2h} + \frac{\delta}{h}\,.$$

Thus, the total error behaves like

$$O(h^\nu) + \frac{\delta}{h}, \tag{1.4}$$

where $\nu = 1$ or 2 if $f \in C^2[0,1]$ or $f \in C^3[0,1]$, respectively. For a fixed error level δ, it looks as in Figure 1.1.

If h becomes too small, the total error increases due to the error term δ/h, the propagated data error. Of course, if h is too large, then the approximation error becomes too large. There is an *optimal* discretization parameter h_0, which can, however, not be computed explicitly, since it depends on unavailable information about the exact data, e.g., their smoothness. However, one can at least estimate the asymptotic behaviour of h_0 if h is chosen as a power of δ, i.e.,

$$h \sim \delta^\mu,$$

then one can minimize (1.4) by taking $\mu = 1/2$ or $\mu = 1/3$, which results in a behaviour of the total error as $O(\sqrt{\delta})$ or $O(\delta^{\frac{2}{3}})$ for $f \in C^2[0,1]$ or $f \in C^3[0,1]$,

Aster et. al 'Parameter Estimation and Inverse Problems'

CHAPTER ONE

Introduction

Synopsis

General issues associated with parameter estimation and inverse problems are introduced through the concepts of the forward problem and its inverse solution. Scaling and superposition properties that characterize linear systems are given, and common situations leading to linear and nonlinear mathematical models are discussed. Examples of discrete and continuous linear and nonlinear parameter estimation problems to be revisited in later chapters are shown. Mathematical demonstrations highlighting the key issues of solution existence, uniqueness, and instability are presented and discussed.

1.1. CLASSIFICATION OF PARAMETER ESTIMATION AND INVERSE PROBLEMS

Scientists and engineers frequently wish to relate physical parameters characterizing a **model**, m, to collected observations making up some set of **data**, d. We will commonly assume that the fundamental physics are adequately understood, so a function, G, may be specified relating m and d such that

$$G(m) = d. \tag{1.1}$$

In practice, d may be a function of time and/or space, or may be a collection of discrete observations. An important issue is that actual observations always contain some amount of noise. Two common ways that noise may arise are unmodeled influences on instrument readings and numerical round-off. We can thus envision data as generally consisting of noiseless observations from a "perfect" experiment, d_{true}, plus a noise component η,

$$d = G(m_{\text{true}}) + \eta \tag{1.2}$$

$$= d_{\text{true}} + \eta, \tag{1.3}$$

where d_{true} exactly satisfies (1.1) for m equal to the true model, m_{true}, and we assume that the forward modeling is exact. We will see that it is commonly mathematically possible, although practically undesirable, to also fit all or part of η by (1.1). It may seem remarkable, but it is often the case that a solution for m that is influenced by even a small noise amplitude η can have little or no correspondence to m_{true}. Another key issue that may seem astounding at first is that commonly there are an infinite number of models aside from m_{true} which fit the perfect data, d_{true}.

When m and d are functions, we typically refer to G as an **operator**. G will be called a function when m and d are vectors. The operator G can take on many forms. In some cases G is an ordinary differential equation (ODE) or partial differential equation (PDE). In other cases, G is a linear or nonlinear system of algebraic equations.

Note that there is some inconsistency between mathematicians and other scientists in modeling terminology. Applied mathematicians usually refer to $G(m) = d$ as the "mathematical model" and to m as the "parameters." On the other hand, scientists often refer to G as the "forward operator" and to m as the "model." We will adopt the scientific parlance and refer to m as the "the model" while referring to the equation $G(m) = d$ as the "mathematical model."

The **forward problem** is to find d given m. Computing $G(m)$ might involve solving an ODE or PDE, evaluating an integral, or applying an algorithm for which there is no explicit analytical formula for $G(m)$. Our focus in this text is on the **inverse problem** of finding m given d. A third problem, not addressed here, is the **model identification problem** of determining G given examples of m and d.

In many cases, we will want to determine a finite number of parameters, n, to define a model. The parameters may define a physical entity directly (e.g., density, voltage, seismic velocity), or may be coefficients or other constants in a functional relationship that describes a physical process. In this case, we can express the model parameters as an n element vector $\mathbf{m}$. Similarly, if there are a finite number of data points then we can express the data as an m element vector $\mathbf{d}$. (Note that the use of the integer m here for the number of data points is easily distinguishable from the model m by its context.) Such problems are called **discrete inverse problems** or **parameter estimation problems**. A general parameter estimation problem can be written as a system of equations

$$G(\mathbf{m}) = \mathbf{d}. \tag{1.4}$$

In other cases, where the model and data are functions of continuous variables, such as time or space, the associated task of estimating m from d is called a **continuous inverse problem**. A central theme of this book is that continuous inverse problems can often be well-approximated by discrete inverse problems.

We will generally refer to problems with small numbers of parameters as "parameter estimation problems." Problems with a larger number of parameters, and which will often require the application of stabilizing constraints, will be referred to as "inverse problems." A key aspect of many inverse problems is that they are ill-conditioned in a sense that will be discussed later in this chapter. In both parameter estimation and inverse problems we solve for a set of parameters that characterize a model, and a key point of this text is that the treatment of all such problems can be sufficiently generalized so

Parameter Estimation and Inverse Problems, Second Edition. DOI: 10.1016/B978-0-12-385048-5.00001-X
© 2013 Elsevier Inc. All rights reserved.

Aster et. al 'Parameter Estimation and Inverse Problems'

that the distinction is largely irrelevant. In practice, what is important is the distinction between ill-conditioned and well-conditioned parameter estimation problems.

A type of mathematical model for which many useful results exist is the class of **linear systems.** Linear systems obey superposition

$$G(m_1 + m_2) = G(m_1) + G(m_2) \tag{1.5}$$

and scaling

$$G(\alpha m) = \alpha G(m). \tag{1.6}$$

In the case of a discrete linear inverse problem, (1.4) can always be written in the form of a linear system of algebraic equations (see Exercise 1.1).

$$G(\mathbf{m}) = \mathbf{Gm} = \mathbf{d}. \tag{1.7}$$

In a continuous linear inverse problem, G can often be expressed as a linear integral operator, where (1.1) has the form

$$\int_a^b g(x, \xi) m(\xi) d\xi = d(x) \tag{1.8}$$

and the function $g(x, \xi)$ is called the **kernel**. The linearity of (1.8) is easily seen because

$$\int_a^b g(x, \xi)(m_1(\xi) + m_2(\xi)) d\xi = \int_a^b g(x, \xi) m_1(\xi) d\xi + \int_a^b g(x, \xi) m_2(x) d\xi \tag{1.9}$$

and

$$\int_a^b g(x, \xi) \alpha m(\xi) d\xi = \alpha \int_a^b g(x, \xi) m(\xi) d\xi. \tag{1.10}$$

Equations in the form of (1.8), where $m(x)$ is the unknown, are called **Fredholm integral equations of the first kind (IFK)**. IFKs arise in a surprisingly large number of inverse problems. A key characteristic of these equations is that they have mathematical properties which make it difficult to obtain useful solutions by straightforward methods.

In many cases the kernel in (1.8) can be written to depend explicitly on $x - \xi$, producing a **convolution equation**,

$$\int_{-\infty}^{\infty} g(x - \xi) m(\xi) d\xi = d(x). \tag{1.11}$$

Here we have written the interval of integration as extending from minus infinity to plus infinity, but other intervals can easily be accommodated by having $g(x - \xi) = 0$ outside of the interval of interest. When a forward problem has the form of (1.11), determining $d(x)$ from $m(x)$ is called **convolution**, and the inverse problem of determining $m(x)$ from $d(x)$ is called **deconvolution**.

Another IFK arises in the problem of inverting a **Fourier transform**

$$\Phi(f) = \int_{-\infty}^{\infty} e^{-\imath 2\pi f x} \phi(x) dx \tag{1.12}$$

(where $\imath$ is $\sqrt{-1}$) to obtain $\phi(x)$. Although there are many tables and analytic methods of obtaining Fourier transforms and their inverses, numerical estimates of $\phi(x)$ may be of interest, such as where there is no analytic inverse, or where we wish to estimate $\phi(x)$ from spectral data collected at discrete frequencies.

It is an intriguing question as to why linearity appears in many interesting physical problems. A principal reason is that many such systems are associated with only small departures from equilibrium. One important geophysical example is seismic wave propagation, where the stresses associated with elastic fields are often very small relative to the elastic moduli of the medium. This situation leads to small strains and to a very nearly linear stress–strain relationship. Because of this, seismic wave field problems in many useful circumstances obey superposition and scaling. Other fields such as gravity and magnetism, at field strengths typically encountered in geophysics, also show effectively linear physics.

Because many important inverse problems are linear, and because linear theory is a key component in solving nonlinear problems, Chapters 2 through 8 of this book cover theory and methods for the solution of linear inverse problems. Nonlinear mathematical models arise when the parameters of interest have an inherently nonlinear relationship to the observables. This situation commonly occurs, for example, in electromagnetic field problems where we wish to relate geometric model parameters such as layer thicknesses to observed field properties. We discuss methods for nonlinear parameter estimation and inverse problems in Chapters 9 and 10, respectively.

1.2. EXAMPLES OF PARAMETER ESTIMATION PROBLEMS

Example 1.1

A canonical parameter estimation problem is the fitting of a function, defined by a collection of parameters, to a data set. In cases where this function fitting procedure can be cast as a linear inverse problem, the procedure is referred to as **linear regression**. An ancient example of linear regression is the characterization of a ballistic trajectory. In a basic take on this problem, the data, **y**, are altitude observations of a ballistic body at a set of times **t**

Aster et. al 'Parameter Estimation and Inverse Problems'

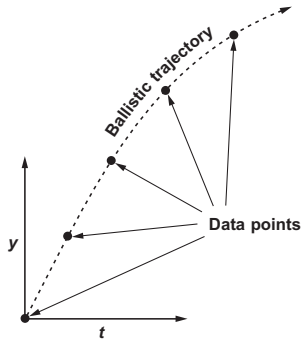


Figure 1.1 The parabolic trajectory problem.

(Figure 1.1). We wish to solve for a model, $\mathbf{m}$, that contains the initial altitude (m_1), initial vertical velocity (m_2), and effective gravitational acceleration (m_3) experienced by the body during its trajectory. This and related problems are naturally of practical interest in aeronautics and weaponry, but are also of fundamental geophysical interest, for example, in absolute gravity meters capable of estimating g from the acceleration of a falling object in a vacuum to accuracies on the order of one part in 10^9 [91].

The mathematical model is a quadratic function in the (t, y) plane

$$y(t) = m_1 + m_2 t - (1/2) m_3 t^2 \tag{1.13}$$

that we expect to apply at all times along the trajectory, not just at the times t_i when we happen to have observations. The data will consist of m observations of the height of the body y_i at corresponding times t_i. Assuming that the t_i are measured precisely, and applying (1.13) to each observation, we obtain a system of equations with m rows and $n = 3$ columns that relates the data y_i to the model parameters, m_j

$$\begin{bmatrix} 1 & t_1 & -\frac{1}{2}t_1^2 \\ 1 & t_2 & -\frac{1}{2}t_2^2 \\ 1 & t_3 & -\frac{1}{2}t_3^2 \\ \vdots & \vdots & \vdots \\ 1 & t_m & -\frac{1}{2}t_m^2 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}. \tag{1.14}$$

Although the mathematical model of (1.13) is quadratic, the equations for the three parameters m_i in (1.14) are linear, so solving for $\mathbf{m} = [m_1, m_2, m_3]^T$ is a linear parameter estimation problem.

If there are more data points than model parameters in (1.14) ($m > n$), then the m constraint equations in (1.14) will likely be inconsistent, and it will be impossible to find

a model $\mathbf{m}$ that satisfies every equation exactly. The nonexistence of a model exactly satisfying the observations in such a case can be seen geometrically because no parabola will exist that goes through all of the (t_i, y_i) points (Exercise 1.2). Such a situation could arise in practice because of noise in the determinations of the y_i, and/or because the forward model of (1.13) is approximate. For example, we have neglected the physics of atmospheric drag, so a true trajectory will not be exactly parabolic and thus exactly modeled by (1.13).

In elementary linear algebra, where an exact solution is typically expected for a system of linear equations, we might throw up our hands at this point and simply state that no solution exists. However, useful solutions to such systems may be found in practice by solving for model parameters that satisfy the data in an approximate, or "best-fit," sense.

A reasonable approach to finding the "best" approximate solution to an inconsistent system of linear equations is to find an $\mathbf{m}$ that minimizes some misfit measure, calculated from the differences between the observations and the theoretical predictions of the forward problem, commonly called **residuals**. A traditional and very widely applied strategy is to find the model that minimizes the **2-norm** (Euclidean length) of the residual vector

$$\|\mathbf{y} - \mathbf{Gm}\|_2 = \sqrt{\sum_{i=1}^{m} (y_i - (\mathbf{Gm})_i)^2}. \tag{1.15}$$

However, (1.15) is not the only, or necessarily the best, misfit measure that can be applied to approximate solve systems of equations. An alternative misfit measure that is superior in many situations is the **1-norm**

$$\|\mathbf{y} - \mathbf{Gm}\|_1 = \sum_{i=1}^{m} |y_i - (\mathbf{Gm})_i|. \tag{1.16}$$

We shall see in Chapter 2 that a solution that minimizes (1.16) is less sensitive to data points that are wildly discordant with the mathematical model than one that minimizes (1.15). Solution techniques that are resistant to such data **outliers** are called **robust estimation procedures**.

Example 1.2

A classic nonlinear parameter estimation problem in geophysics is determining the space and time coordinates of an earthquake nucleation, the hypocenter, which is specified by

Aster et. al 'Parameter Estimation and Inverse Problems'

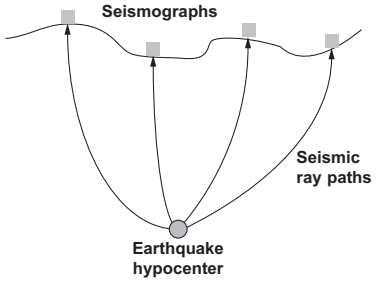


Figure 1.2 The earthquake location problem.

the four-vector

$$\mathbf{m} = \begin{bmatrix} \boldsymbol{\xi} \\ \tau \end{bmatrix} \tag{1.17}$$

where $\boldsymbol{\xi}$ is the three-dimensional earthquake location and the fourth element, τ, is the earthquake origin time (Figure 1.2). The hypocenter model we seek best fits a vector of seismic phase **arrival times**, $\mathbf{t}$, observed at an m-station seismic network. The mathematical model is

$$G(\mathbf{m}) = \mathbf{t} \tag{1.18}$$

where G models the physics of seismic wave propagation to map a hypocenter into a vector of predicted seismic arrival times at m stations. G depends on the seismic velocity structure, $v(\mathbf{x})$, which we assume here to be known.

The earthquake location problem is nonlinear even if $v(\mathbf{x})$ is a constant, c. In this case, all of the ray paths in Figure 1.2 would be straight, and the arrival time of the seismic signal at station i would be

$$t_i = \frac{\|\mathbf{S}_{\cdot,i} - \boldsymbol{\xi}\|_2}{c} + \tau \tag{1.19}$$

where the ith column of the matrix $\mathbf{S}$, $\mathbf{S}_{\cdot,i}$, specifies the coordinates for station i. Equation (1.19) is nonlinear with respect to the spatial parameters ξ_i in $\mathbf{m}$, and thus the problem cannot be expressed as a linear system of equations.

In a few special cases, a change of variables can convert a nonlinear problem to a linear one. More generally, nonlinear parameter estimation problems can often be solved by choosing a starting model and then iteratively improving it until a good solution is obtained. General methods for solving nonlinear parameter estimation problems are discussed in Chapter 9.

1.3. EXAMPLES OF INVERSE PROBLEMS

Example 1.3

In **vertical seismic profiling**, we wish to know the vertical seismic velocity of the material surrounding a borehole. A downward-propagating seismic wavefront is generated at the surface by a source, and seismic waves are sensed by a string of seismometers in the borehole (Figure 1.3).

The arrival times of the seismic wavefront at each instrument, which reflects the seismic velocity for vertically traveling waves as a function of depth, are measured from the recorded seismograms. The problem is nonlinear if it is expressed in terms of seismic velocities. However, we can linearize this problem via a simple change of variables where we parameterize the seismic structure in terms of **slowness**, $s(z)$, the reciprocal of the velocity $v(z)$. The observed travel time at depth z can then be expressed as the definite integral of the vertical slowness, s, from the surface to z:

$$t(z) = \int_0^z s(\xi)\,d\xi \tag{1.20}$$

$$= \int_0^\infty s(\xi) H(z-\xi)\,d\xi \tag{1.21}$$

where the kernel function H is the **Heaviside step function**, which is equal to one when its argument is nonnegative and zero when its argument is negative. The explicit dependence of the kernel on $z-\xi$ shows that (1.21) is a convolution.

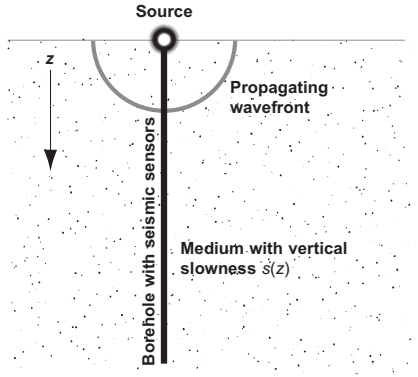


Figure 1.3 The vertical seismic profiling problem.

Aster et. al 'Parameter Estimation and Inverse Problems'

In theory, we can solve (1.21) quite easily because, by the fundamental theorem of calculus,

$$s(z) = \frac{dt(z)}{dz}. \tag{1.22}$$

In practice, simply differentiating the observations to obtain a solution may not be useful. This is because there will generally be noise present in the observed times $t(z)$, and applying (1.22) may result in a solution that includes unphysical values of $s(z)$ or other unrealistic model features.

Example 1.4

A further instructive linear inverse problem is the inversion of a vertical gravity anomaly, $d(x)$, observed at some height, h, to estimate an unknown buried line mass density distribution deviation from a background model, $m(x) = \Delta\rho(x)$ (Figure 1.4). The mathematical model for this problem can be written as an IFK, because the data are a superposition of the vertical gravity contributions from the differential elements comprising the line mass

$$d(x) = \Gamma \int_{-\infty}^{\infty} \frac{h}{\left((\xi - x)^2 + h^2\right)^{3/2}} m(\xi)\, d\xi \tag{1.23}$$

$$= \int_{-\infty}^{\infty} g(\xi - x) m(\xi)\, d\xi \tag{1.24}$$

where Γ is Newton's gravitational constant. Note that the kernel has the form $g(\xi - x)$, and (1.24) is thus a convolution. Because the kernel is a smooth function, $d(x)$ will be a smoothed and scaled transformation of $m(x)$. Conversely, solutions for $m(x)$ will be a roughened transformation of $d(x)$. For this reason we again need to be wary of possibly severe deleterious effects of noise in the data.

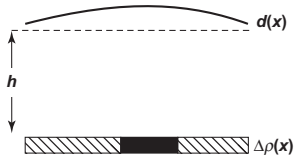


Figure 1.4 A linear inverse problem: determine the density deviation of a buried line mass, $\Delta\rho(x)$, relative to a background model, from gravity anomaly observations $d(x)$.

Example 1.5

Consider a variation on Example 1.4, where the depth of the line density perturbation varies, rather than the density contrast. The gravity anomaly is now attributable to a variation in the burial depth, $m(x) = h(x)$, of an assumed known density perturbation, $\Delta\rho$ (Figure 1.5). The physics is the same as in Example 1.4, so the data are still given by the superposition of density perturbation contributions to the gravitational anomaly field, but the mathematical model now takes the form

$$d(x) = \Gamma \int_{-\infty}^{\infty} \frac{m(\xi)}{\left((\xi - x)^2 + m^2(\xi)\right)^{3/2}} \Delta\rho\, d\xi. \tag{1.25}$$

This problem is nonlinear in $m(x)$ because (1.25) does not follow the superposition and scaling rules (1.5) and (1.6).

Nonlinear inverse problems are generally significantly more difficult to solve than linear ones. In special cases, they may be solvable by coordinate transformations that globally linearize the problem or by other clever special–case methods. In other cases, the problem cannot be globally linearized, so nonlinear optimization techniques must be applied. The essential differences in the treatment of linear and nonlinear problems arise because, as we shall see, all linear problems can be generalized to be the "same" in an important sense, so that a single set of solution methods can be applied to all. In contrast, nonlinear problems tend to be nonlinear in mathematically different ways and

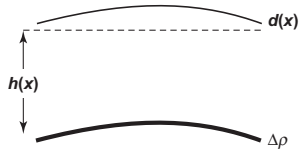


Figure 1.5 A nonlinear inverse problem: determine the depth to a buried line mass density anomaly $h(x)$ from observed gravity anomaly observations $d(x)$.

Aster et. al 'Parameter Estimation and Inverse Problems'

either require specific strategies or, more commonly, can by solved by iterative methods that may rely on local linearization.

Example 1.6

A classic pedagogical inverse problem is an experiment in which an angular distribution of illumination passes through a thin slit and produces a diffraction pattern, for which the intensity is observed (Figure 1.6; [141]).

The data, $d(s)$, are measurements of diffracted light intensity as a function of the outgoing angle $-\pi/2 \le s \le \pi/2$. Our goal is to find the intensity of the incident light on the slit, $m(\theta)$, as a function of the incoming angle $-\pi/2 \le \theta \le \pi/2$.

The forward problem relating d and m can be expressed as the linear mathematical model,

$$d(s) = \int_{-\pi/2}^{\pi/2} (\cos(s) + \cos(\theta))^2 \left(\frac{\sin(\pi(\sin(s) + \sin(\theta)))}{\pi(\sin(s) + \sin(\theta))} \right)^2 m(\theta) d\theta. \tag{1.26}$$

Example 1.7

Consider the problem of recovering the history of groundwater pollution at a source site from later measurements of the contamination at downstream wells to which the contaminant plume has been transported by advection and diffusion (Figure 1.7). This

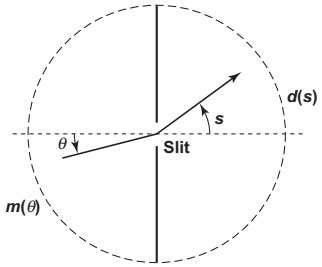


Figure 1.6 The Shaw diffraction intensity problem (1.26).

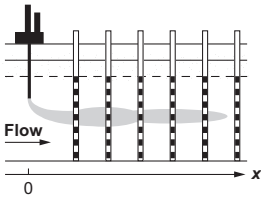


Figure 1.7 The contaminant plume source history reconstruction problem.

"source history reconstruction problem" has been considered by a number of authors [117, 144, 145].

The mathematical model for contaminant transport is an advection–diffusion equation

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2} - v\frac{\partial C}{\partial x} \tag{1.27}$$

$$C(0,\ t) = C_{in}(t)$$

$$C(x,\ t) \to 0 \ \text{ as } \ x \to \infty$$

where D is the diffusion coefficient and v is the velocity of groundwater flow. The solution to (1.27) at time T is the convolution

$$C(x,\ T) = \int_0^T C_{in}(t) f(x,\ T-t)\, dt, \tag{1.28}$$

where $C_{in}(t)$ is the time history of contaminant injection at $x = 0$, and the kernel is

$$f(x,\ T-t) = \frac{x}{2\sqrt{\pi D (T-t)^3}} e^{-\frac{[x - v(T-t)]^2}{4D(T-t)}}. \tag{1.29}$$

Example 1.8

An important and instructive inverse problem is **tomography**, from the Greek roots *tomos*, "to section" or "to cut" (the ancient concept of an *atom* was that of an irreducible, uncuttable object) and *graphein*, "to write." Tomography is the general technique of determining models that are consistent with path-integrated properties such as attenuation (e.g., X-ray, radar, muon, seismic), travel time (e.g., electromagnetic, seismic, or

Aster et. al 'Parameter Estimation and Inverse Problems'

acoustic), or source intensity (e.g., positron emission). Here, we will use examples from seismic tomography. Although tomography problems originally involved determining models that were two-dimensional slices of three-dimensional objects, the term is now commonly used in situations where the model is two- or three-dimensional. Tomography has many applications in medicine, engineering, acoustics, and Earth science. One important geophysical example is cross-well tomography, where seismic sources are installed in a borehole, and the signals are received by sensors in another borehole. Another example is joint earthquake location/velocity structure inversion carried out on scales ranging from a fraction of a cubic kilometer to global [158, 159, 160].

The physical model for tomography in its most basic form (Figure 1.8) assumes that geometric ray theory (essentially the high-frequency limiting case of the wave equation) is valid, so that wave energy traveling between a source and receiver can be considered to be propagating along infinitesimally narrow ray paths. The density of ray path coverage in a tomographic problem may vary significantly throughout a section or volume, and provide much better constraints on physical properties in densely sampled regions than in sparsely sampled ones.

In seismic tomography, if the slowness at a point $\mathbf{x}$ is $s(\mathbf{x})$, and the ray path ℓ is known, then the travel time for seismic energy transiting along that ray path is given by the line integral along ℓ

$$t = \int_{\ell} s(\mathbf{x}(l))\,dl. \tag{1.30}$$

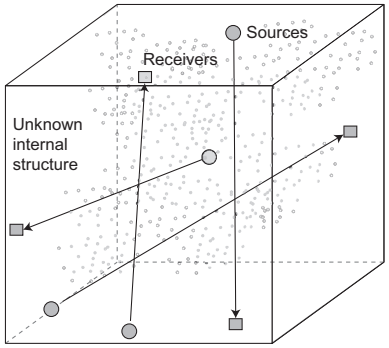


Figure 1.8 Conceptual depiction of ray path tomography. Sources and receivers may, in general, be either at the edges or within the volume, and ray paths may be either straight, as depicted, or curved.

Note that (1.30) is just a higher-dimensional generalization of (1.21), the forward problem for the vertical seismic profiling example. In general, seismic ray paths will be bent due to refraction and/or reflection. In cases where such effects are negligible, ray paths can be usefully approximated as straight lines (e.g., as depicted in Figure 1.8), and the forward and inverse problems can be cast in a linear form. However, if the ray paths are significantly bent by slowness variations, the resulting inverse problem will be nonlinear.

1.4. DISCRETIZING INTEGRAL EQUATIONS

Consider problems of the form

$$d(x) = \int_a^b g(x, \xi)m(\xi)d\xi. \tag{1.31}$$

Here $d(x)$ is a known function, typically representing observed data. The kernel $g(x, \xi)$ is considered to be given, and encodes the physics relating an unknown model $m(x)$ to observed data $d(s)$. The interval $[a, b]$ may be finite or infinite. The function $d(x)$ might in theory be known over an entire interval, but in practice we will only have measurements of $d(x)$ at a finite set of points.

We wish to solve for $m(x)$. This type of linear equation is called a Fredholm integral equation of the first kind, or IFK. A surprisingly large number of inverse problems, including all of the examples from the previous section, can be written as IFKs. Unfortunately, IFKs have properties that can make them very challenging to solve.

To obtain useful numerical solutions to IFKs, we will frequently discretize them into forms that are tractably solvable using the methods of linear algebra. We first assume that $d(x)$ is known at a finite number of points $x_1, x_2, \ldots, x_m$. We can then write the forward problem as

$$d_i = d(x_i) = \int_a^b g(x_i, \xi)m(\xi)d\xi \quad i = 1, 2, \ldots, m \tag{1.32}$$

or as

$$d_i = \int_a^b g_i(x)m(x)dx \quad i = 1, 2, \ldots, m, \tag{1.33}$$

where $g_i(x) = g(x_i, \xi)$. The functions $g_i(t)$ are referred to as **representers** or **data kernels**.

Aster et. al 'Parameter Estimation and Inverse Problems'

Here, we will apply a **quadrature rule** to approximate (1.33) numerically. Note that, although quadrature methods are applied in this section to linear integral equations, they also have utility in the discretization of nonlinear problems. The simplest quadrature approach is the **midpoint rule**, where we divide the interval $[a, b]$ into n subintervals, and pick points $x_1, x_2, \ldots, x_n$ in the middle of each interval. The points are given by

$$x_j = a + \frac{\Delta x}{2} + (j-1)\Delta x \tag{1.34}$$

where

$$\Delta x = \frac{b-a}{n}. \tag{1.35}$$

The integral (1.33) is then approximated by (Figure 1.9):

$$d_i = \int_a^b g_i(x)m(x)dx \approx \sum_{j=1}^{n} g_i(x_j)m(x_j)\Delta x, \quad i = 1, 2, \ldots, m. \tag{1.36}$$

If we let

$$G_{i,j} = g_i(x_j)\Delta x \quad \begin{pmatrix} i = 1, 2, \ldots, m \\ j = 1, 2, \ldots, n \end{pmatrix} \tag{1.37}$$

and

$$m_j = m(x_j) \quad j = 1, 2, \ldots, n, \tag{1.38}$$

then we obtain a linear system of equations $\mathbf{Gm} = \mathbf{d}$.

The approach of using the midpoint rule to approximate the integral is known as **simple collocation**. Of course, there are also more sophisticated quadrature rules for numerically approximating integrals (e.g., the trapezoidal rule, or Simpson's rule). In each case, we end up with an m by n linear system of equations, but the formulas for evaluating the elements of $\mathbf{G}$ will be different.

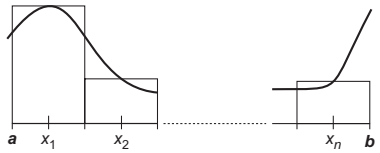


Figure 1.9 Grid for the midpoint rule.

Example 1.9

Consider vertical seismic profiling (Example 1.3), where we wish to estimate vertical seismic slowness using travel time measurements of downward propagating seismic waves (Figure 1.10). We discretize the forward problem (1.21) for m travel time data points, t_i, and n model depths, z_j, spaced at constant intervals of Δz. This discretization is depicted in Figure 1.10.

The discretized problem has

$$t_i = \sum_{j=1}^{n} H(y_i - z_j)s_j\Delta z, \tag{1.39}$$

where $n/m = \Delta y/\Delta z$ is an integer. The rows, $\mathbf{G}_{i,\cdot}$, of the matrix each consist of $i \cdot n/m$ elements equal to Δz on the left and $n - (i \cdot n/m)$ zeros on the right. For $n = m$, $\mathbf{G}$ is a lower triangular matrix with each nonzero entry equal to Δz.

Example 1.10

To discretize the Shaw problem (1.26), we apply the method of simple collocation with m and n intervals for the data and model functions, respectively. We additionally define

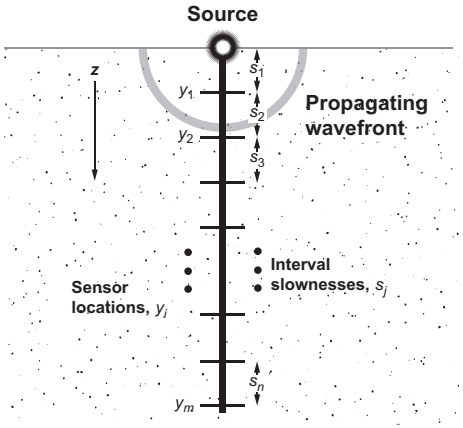


Figure 1.10 Discretization of the vertical seismic profiling problem ($n = 2m$) into uniform intervals.

Aster et. al 'Parameter Estimation and Inverse Problems'

the data and model points at m and n equally spaced respective angles given by

$$s_i = \frac{(i-0.5)\pi}{m} - \frac{\pi}{2} \quad i = 1,\ 2,\ \ldots,\ m \tag{1.40}$$

and

$$\theta_j = \frac{(j-0.5)\pi}{n} - \frac{\pi}{2} \quad j = 1,\ 2,\ \ldots,\ n. \tag{1.41}$$

Correspondingly, discretizing the data and model into m- and n-length vectors

$$d_i = d(s_i) \quad i = 1,\ 2,\ \ldots,\ m \tag{1.42}$$

and

$$m_j = m(\theta_j) \quad j = 1,\ 2,\ \ldots,\ n \tag{1.43}$$

leads to a discrete linear system $\mathbf{Gm} = \mathbf{d}$, where

$$G_{i,j} = (\cos(s_i) + \cos(\theta_j))^2 \left(\frac{\sin(\pi(\sin(s_i) + \sin(\theta_j)))}{\pi(\sin(s_i) + \sin(\theta_j))} \right)^2 \Delta\theta \tag{1.44}$$

and

$$\Delta\theta = \frac{\pi}{n}. \tag{1.45}$$

Example 1.11

We discretize the advection–diffusion problem (1.27), assuming that the parameters D and v in (1.29) are known. We wish to estimate $C_{in}(t)$ from simultaneous concentration observations at the locations x_i at some later time T. The convolution (1.28) for $C(x,\ T)$ is discretized as

$$\mathbf{d} = \mathbf{Gm} \tag{1.46}$$

where $\mathbf{d}$ is a vector of sampled concentrations at different well locations, $\mathbf{x}$, at time T, $\mathbf{m}$ is a vector of C_{in} values to be estimated, and

$$G_{i,j} = f(x_i,\ T - t_j)\Delta t \tag{1.47}$$

$$= \frac{x_i}{2\sqrt{\pi D(T-t_j)^3}} e^{-\frac{[x_i - v(T-t_j)]^2}{4D(T-t_j)}} \Delta t. \tag{1.48}$$

Example 1.12

A common way of discretizing the model in a tomographic problem is as uniform blocks (Figure 1.11). This approach is essentially applying the midpoint rule to the travel-time forward problem (1.30).

In this parameterization, the elements of $\mathbf{G}$ are just the lengths of the ray paths within corresponding blocks. Consider the example of Figure 1.11, where nine homogeneous blocks with sides of unit length and unknown slowness are crossed by eight ray paths. For straight ray paths, we map the two-dimensional slowness grid to a model vector using a row-by-row indexing convention. The constraint equations in the mathematical model are then

$$\mathbf{Gm} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ \sqrt{2} & 0 & 0 & 0 & \sqrt{2} & 0 & 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} s_{1,1} \\ s_{1,2} \\ s_{1,3} \\ s_{2,1} \\ s_{2,2} \\ s_{2,3} \\ s_{3,1} \\ s_{3,2} \\ s_{3,3} \end{bmatrix} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \\ t_6 \\ t_7 \\ t_8 \end{bmatrix}. \tag{1.49}$$

Because there are nine unknown parameters $s_{i,j}$ in the model, but only eight constraints, the $\mathbf{G}$ matrix is clearly rank deficient. In fact, rank($\mathbf{G}$) is only seven. In addition,

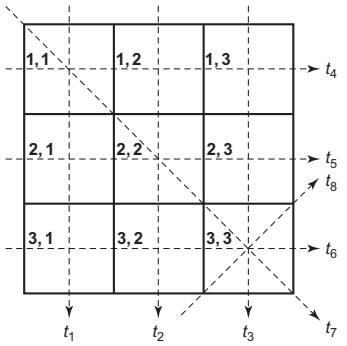


Figure 1.11 Discretization of a tomography problem into uniform blocks. Ray paths correspond to the constraint equations in (1.49).

Aster et. al 'Parameter Estimation and Inverse Problems'

there is clearly redundant information in (1.49), in that the slowness $s_{3,3}$ is completely determined by t_8, yet $s_{3,3}$ also influences the observations t_3, t_6, and t_7.

1.5. WHY INVERSE PROBLEMS ARE DIFFICULT

Scientists and engineers need to be concerned with far more than simply finding mathematically acceptable answers to parameter estimation and inverse problems. One reason is that there may be many models that adequately fit the data. It is essential to characterize just what solution has been obtained, how "good" it is in terms of physical plausibility, its ability to predict the data, and perhaps how consistent it is with other constraints. Essential issues that must be considered include **solution existence**, **solution uniqueness**, and **instability** of the solution process.

1. *Existence*. There may be no model that exactly fits the data. This can occur in practice because the mathematical model of the system's physics is approximate (or perhaps simply incorrect) or because the data contain noise.
2. *Uniqueness*. If exact solutions do exist, they may not be unique, even for an infinite number of exact data points. That is, there may be other solutions besides m_{true} that exactly satisfy $G(m) = d_{\text{true}}$. This situation commonly occurs in potential field problems. A classic example is the external gravitational field from a spherically symmetric mass distribution, which depends only on the total mass, and not on the radial density distribution.

   Nonuniqueness is a characteristic of rank deficient discrete linear inverse problems because the matrix $\mathbf{G}$ in this case has a nontrivial **null space**. In linear inverse problems, models ($\mathbf{m}_0$) that lie in the null space of $\mathbf{G}$ are solutions to $\mathbf{G}\mathbf{m}_0 = \mathbf{0}$. By superposition, any linear combination of these **null space models** can be added to a particular model that satisfies (1.7) and not change the fit to the data. There are thus an infinite number of mathematically acceptable models in such situations. In practical terms, suppose that there exists a nonzero model $\mathbf{m}_0$ that results in an instrument reading of zero. We cannot discriminate this situation from the situation where $\mathbf{m}_0$ is truly zero.

   An important and thorny issue with problems that have nonunique solutions is that an estimated model may be significantly smoothed or otherwise biased relative to the true situation. Characterizing such bias is essential to interpreting models in terms of their possible correspondence to reality. This issue falls under the general topic of **model resolution analysis**.
3. *Instability*. The process of computing an inverse solution can be, and often is, extremely unstable in that a small change in measurement (e.g., a small η in (1.3)) can lead to an enormous change in the estimated model. Inverse problems where this

situation arises are referred to as **ill-posed** in the case of continuous systems, or **ill-conditioned** in the case of discrete linear systems. A key point is that it is commonly possible to stabilize the inversion process by imposing additional constraints that bias the solution, a process that is generally referred to as **regularization**. Regularization is frequently essential to producing a usable solution to an otherwise intractable ill-posed or ill-conditioned inverse problem.

To examine existence, uniqueness, and instability issues, let us consider some simple mathematical examples using an IFK,

$$\int_0^1 g(x, \xi) m(\xi) d\xi = y(x). \tag{1.50}$$

First, consider the trivial case where the kernel is a constant, such as

$$g(x, \xi) = 1, \tag{1.51}$$

which produces the integral equation

$$\int_0^1 m(\xi) d\xi = y(x). \tag{1.52}$$

Because the left-hand side of (1.52) is independent of x, this system has no solution unless $y(x)$ is a constant. Thus, there are an infinite number of mathematically conceivable data sets $y(x)$ that are not constant and for which no exact solution exists. This is a simple illustration of a solution existence issue.

Where a solution to (1.52) does exist, the solution is nonunique because there are an infinite number of functions that, when integrated over the unit interval, produce the same constant and thus satisfy the IFK exactly. This demonstrates a uniqueness issue.

A more subtle example of nonuniqueness can be seen for

$$g(x, \xi) = x \cdot \sin(\pi \xi) \tag{1.53}$$

in (1.50), so that the IFK becomes

$$\int_0^1 x \cdot \sin(\pi \xi) m(\xi) d\xi = y(x). \tag{1.54}$$

Aster et. al 'Parameter Estimation and Inverse Problems'

The functions $\sin(k\pi x)$ for integer values of k are orthogonal in the sense that

$$\int_0^1 \sin(k\pi x)\,\sin(l\pi x)dx = -\frac{1}{2}\int_0^1 \cos(\pi(k+l)x) - \cos(\pi(k-l)x)dx =$$

$$-\frac{1}{2\pi}\left(\frac{\sin(\pi(k+l))}{k+l} - \frac{\sin(\pi(k-l))}{k-l}\right) = 0 \quad (k \neq \pm l;\ k,\ l \neq 0). \tag{1.55}$$

Thus, in (1.54), for models of the form $m(x) = \sin(k\pi x)$, for $k = \pm 2, \pm 3, \ldots$, we have

$$\int_0^1 g(x,\,\xi)m(\xi)d\xi = \int_0^1 g(x,\,\xi)\,\sin(k\pi\xi)d\xi = 0. \tag{1.56}$$

Furthermore, because (1.54) is a linear system, we can add any function of the form

$$m_0(x) = \sum_{k=2}^{\infty} \alpha_k\,\sin(k\pi x) \tag{1.57}$$

to a solution, $m(x)$, and obtain a new model that fits the data equally well.

$$\int_0^1 x\cdot\sin(\pi\xi)(m(\xi)+m_0(\xi))d\xi = \int_0^1 x\cdot\sin(\pi\xi)m(\xi)d\xi + \int_0^1 s\cdot\sin(\pi\xi)m_0(\xi)d\xi$$

$$= \int_0^1 x\cdot\sin(\pi\xi)m(\xi)d\xi + 0. \tag{1.58}$$

There are thus an infinite number of very different solutions that fit the data equally well.

Finally, even if we do not encounter existence or uniqueness issues, instability is a fundamental feature of IFKs. It can be shown that, in the limit as k goes to infinity,

$$\lim_{k\to\infty}\int_{-\infty}^{\infty} g(x,\,\xi)\,\sin\,k\pi\xi\,d\xi = 0 \tag{1.59}$$

for *all* square-integrable functions $g(x,\,\xi)$. This result is known as the **Riemann-Lebesgue lemma** [134]. Examining (1.59) in more detail, we can better understand why this occurs. The oscillatory sine function is smoothed by integration with the kernel $g(\xi,\,x)$. For sufficiently large values of the sine frequency, k, the positive and negative excursions of the sine function will average out to zero. The inverse problem has this situation reversed, so that an inferred model can be extremely sensitive to small changes

in the data. If such small data changes are created by random noise that has nothing to do with the physical system that we are studying, then an inferred model from solving an inverse problem to fit this noise can easily have essentially no correspondence to the true model.

The unstable character of IFK solutions is similar to the situation encountered in solving linear systems of equations where the condition number of the matrix is very large, or equivalently, where the matrix is nearly singular. In both cases, the difficulty lies in the mathematical model itself, and not in the particular algorithm used to solve the problem. Fundamentally, most inverse problems are ill-posed.

1.6. EXERCISES

1. Consider a mathematical model of the form $G(\mathbf{m}) = \mathbf{d}$, where $\mathbf{m}$ is a vector of length n, and $\mathbf{d}$ is a vector of length m. Suppose that the model obeys the superposition and scaling laws and is thus linear. Show that $G(\mathbf{m})$ can be written in the form

$$G(\mathbf{m}) = \mathbf{\Gamma}\mathbf{m} \tag{1.60}$$

where $\mathbf{\Gamma}$ is an m by n matrix. What are the elements of $\mathbf{\Gamma}$? Hint: Consider the standard basis, and write $\mathbf{m}$ as a linear combination of the vectors in the standard basis. Apply the superposition and scaling laws. Finally, recall the definition of matrix-vector multiplication.

2. Can (1.14) be inconsistent, even with only $m = 3$ data points? How about just $m = 2$ data points? If the system can be inconsistent, give an example. If not, explain why.
3. Consider the borehole vertical seismic profile problem of Examples 1.3 and 1.9 for $n = 100$ equally spaced seismic sensors located at depths of $z = 0.2,\ 0.4,\ \ldots,\ 20$ m, and for a model $\mathbf{m}$ describing n corresponding equal-length seismic slowness values for 0.2 m intervals having midpoints at $z - 0.1$ m.
   a. Calculate the appropriate system matrix, $\mathbf{G}$, for discretizing the integral equation (1.21) using the midpoint rule.
   b. For a seismic velocity model having a linear depth gradient specified by

$$v = v_0 + kz, \tag{1.61}$$

   where the velocity at $z = 0$ is $v_0 = 1$ km/s and the gradient is $k = 40$ m/s per m, calculate the true slowness values, $\mathbf{m}_{\text{true}}$, at the midpoints of the n intervals. Integrate the corresponding slowness function for (1.61) using (1.21) to calculate a noiseless synthetic data vector, $\mathbf{d}$, of predicted seismic travel times at the sensor depths.
   c. Solve for the slowness, $\mathbf{m}$, as a function of depth using your $\mathbf{G}$ matrix and analytically calculated noiseless travel times using the MATLAB backslash operator. Compare your result graphically with $\mathbf{m}_{\text{true}}$.

Aster et. al 'Parameter Estimation and Inverse Problems'

d. Generate a noisy travel time vector where independent normally distributed noise with a standard deviation of 0.05 ms is added to the elements of $\mathbf{d}$. Resolve the system for $\mathbf{m}$ and again compare your result graphically with $\mathbf{m}_{\text{true}}$. How has the model changed?

e. Repeat the problem, but for just $n = 4$ sensor depths and corresponding equal length slowness intervals. Is the recovery of the true model improved? Explain in terms of the condition numbers of your $\mathbf{G}$ matrices.

4. Find a journal article that discusses the solution of an inverse problem in a discipline of special interest to you. What are the data? Are the data discrete or continuous? Have the authors discussed possible sources of noise in the data? What is the model? Is the model continuous or discrete? What physical laws determine the forward operator G? Is G linear or nonlinear? Do the authors discuss any issues associated with existence, uniqueness, or instability of solutions?

1.7. NOTES AND FURTHER READING

Some important references on inverse problems in geophysics and remote sensing include [28, 60, 99, 126, 166]. Instructive examples of ill-posed problems and their solutions can be found in the book edited by Tikhonov and Goncharsky [162]. More mathematically oriented references on inverse problems include [8, 45, 59, 65, 67, 90, 94, 108, 112, 155, 161]. Tomography, particularly in medical imaging and seismology, is a very large field. Some general references on tomography are [70, 79, 84, 98, 100, 114, 121].